

Grading, Drainage and the Water Table

Almost every site defect is one question: where does the water go? Your most-missed question on this topic is here.

Drainage issues are the primary source of problems found during a site inspection, and improper lot grading is the most common cause of wet basements. If you learn one thing about the site, learn that the ground must move water away from the building.

Grading	The arrangement and preparation of the soil for construction, placing drainage areas where appropriate for water runoff.
Surface runoff	The loss of water from an area by its flow over the land's surface.
Swale	A shallow depression in the ground used to redirect surface water runoff.
Negative grade	Ground adjacent to the house that slopes toward the foundation instead of away. This is the basic drainage problem that creates water seepage into a basement or crawl space.

The numbers here

Grade must fall away from the building

6 in over the first 10 ft

Impervious surface within 10 ft of the foundation — minimum slope

2 percent

YOU HAVE MISSED THIS 3 TIMES

The loss of water from an area by its flow over the land's surface is known as:

Surface runoff.

YOU HAVE MISSED THIS 2 TIMES

The arrangement and preparation of the soil for construction by placing drainage areas where appropriate for water runoff is called:

Grading.

When the water is already below the house

Grading handles water arriving at the surface. A high water table is water already standing in the ground, pushing up and sideways against the foundation, and grading cannot touch it. That needs a system that collects water and carries it away.

YOU HAVE MISSED THIS 5 TIMES

To counteract problems associated with a high water table, which of the following is used?

Drain tiles.

This is the single most-missed question you own in this topic. Drain tiles are perforated pipe that collects groundwater and moves it somewhere else. Gravel alone lets water travel but never removes it — a drainage system needs somewhere for the water to go, not just somewhere to sit.

Reading the evidence

Evidence of past leakage	Staining, efflorescence, rust marks, damaged stored goods, a tide line on the wall.
Not evidence	Small cracks in the basement floor. A slab cracks from shrinkage and settlement as a matter of course, so a small floor crack tells you nothing about whether water has been in the room.
Trees	Report limbs hanging over the house. You are reporting the hazard to the roof and the gutters, not diagnosing the tree.

Say it back

Cover the page. Answer each one out loud.

1. What is the primary source of problems found on a site inspection?
2. What is the most common cause of wet basements?
3. Give the grading numbers: the fall over the first ten feet, and the slope for paving.
4. Define surface runoff, grading and a swale.
5. What counteracts a high water table, and why is gravel alone not enough?
6. Which basement-floor observation is NOT evidence of previous leakage?
7. What do you write about trees on the property?